∴ The condition that all leading entries in a row have only zeros below them in their respective columns is True.

4. The condition that all leading entries of rows are 1 is True.

5. Row 1 Column 2 = 0
Row 1 Column 4 = Row 2 Column 4 = 0

Using condition 3, we conclude that the condition each leading 1 is the only nonzero entry in its column is True.

$\boxed{\text{The matrix is in reduced echelon form}}$ ⌗

c) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ 1. Third row is a zero row but the fourth row is non-zero. ∴ The condition that every non-zero row must be above all zero rows is False

$\boxed{\therefore \text{The matrix is not in echelon form}}$ ⌗

d) $\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 & 2 \\ 0 & 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$ 1. Each row is non-zero. ∴ The condition that every non-zero row must be above all zero rows is vacuously true.

2. Leading element of row 1 = 1 of Column 1
Leading element of row 2 = 2 of Column 2
Leading element of row 3 = 3 of Column 4
Leading element of row 4 = 4 of Column 5

∴ The condition that the leading element of a row is in a column to the right of the column of the leading element of any row above it is valid.

3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
Row 3 Column 2 = Row 4 Column 2 = 0
Row 4 Column 4 = 0

∴ The condition that all leading entries in a row have only zeros below them in their respective columns is True.

4. The condition that all leading entries must be 1 is False.
We have seen that for each of row 2 to row 4, the leading entry is not 1.